

SSDs Move Ahead to Overtake Hard Disk Drives

High speed, durable and sleek SSDs are moving in to replace 'traditional' HDDs.

A solid state drive (SSD) is a data storage device that uses integrated circuit assemblies as its memory to store data. Now that everyone is switching over to thin tablets and high performance notebooks, carrying heavy, bulky hard disks may be difficult. SSDs, therefore, play a vital role in today's world as they combine high speed, durability and smaller sizes, with vast storage and power efficiency.

SSDs consume minimal power because they do not have any movable parts inside, which leads to less consumption of internal power.

HDDs vs SSDs

The new technologies embedded in SSDs make them costlier than HDDs. "SSDs, with their new technology, will gradually overtake hard disk drives (HDDs), which have been around ever since PCs came into prominence. It takes time for a new technology to completely take over the traditional one. Also, new technologies are usually expensive. However, users are ready to pay a little more for a new technology because it offers better performance," explains Rajesh Gupta, country head and director, Sandisk Corporation India.

SSDs use integrated circuit assemblies as memory for storing data. The technology uses an electronic interface which is compatible with traditional block input/output HDDs. So SSDs can easily replace HDDs in commonly used applications.

An SSD uses a flash-based medium for storage. It is believed to have a longer life than an HDD and also consumes less power. "SSDs are the next stage in the evolution of PC storage. They run faster, and are quieter and cooler than the aging technology inside hard drives. With no moving parts, SSDs are also more durable and reliable than hard drives. They not only boost the performance but can also be used to breathe new life into older systems," says Vishal Parekh, marketing director, Kingston Technology India.

How to select the right SSD

If you're a videographer, or have a studio dedicated to audio/video post-production work, or are in the banking sector, you can look at ADATA's latest launch, which has been featured later in the article. Kingston, too, has introduced SSDs for all possible purposes. SSDs are great options even for gamers, or those who want to ensure their data has been saved in a secure

medium. Kingston offers an entire range of SSDs, including entry level variants as well as options for general use.

There are a lot of factors to keep in mind when you are planning to buy an SSD—durability, portability, power consumption and speed. Gupta adds that, "The performance of SSDs is typically indicated by their IOPS (Input output operation per second), so one should look at the specifications of the product. Also, check the storage capacity. If you're looking for an SSD when you already have a PC or laptop, then double check the compatibility between your system and the SSD you've shortlisted. If you're buying a new system, then you can always check with the vendors as to what SSD options are available. Research the I/O speeds and get updates about how reliable the product is."

"For PC users, some of the important performance parameters of SSDs are related to battery life, heating of the device and portability. An SSD is 100 per cent solid state technology and has no motor inside, so the advantage is that it consumes less energy; hence, it extends the battery life of the device and is quite portable," explains Gupta.

Listed below are a few broad specifications of SSDs, which can help buyers decide which variant to go in for.

Portability

Portability is one of the major concerns when buying an external hard drive because, as discussed earlier, everyone is gradually shifting to tablets, iPads and notebooks and so would not want to carry around an external hard disk that is heavier than the computing device. The overall portability of an SSD is evaluated on the basis of its size, shape, how much it weighs and its ruggedness.

High speed

Speed is another factor people look for, while buying an SSD. If it is not fast, it is not worth the buy. SSDs offer data transfer read speeds that range from approximately 530 MBps to 550 MBps, whereas a HDD offers only around 30 to 50 MBps. SSDs can also boot any operating system almost four times faster than a traditional 7200 RPM 500 GB hard drive disk. With SSDs, the applications provide a 12 times faster response compared to the HDD. A system equipped with an SSD also launches applications faster and offers a high performance overall.